

The Source Code

Open a new file in your editor or IDE, and enter the following code. Save it as *towerofhanoi.py*.

```
"""THE TOWER OF HANOI, by Al Sweigart al@inventwithpython.com
A stack-moving puzzle game."""

import copy
import sys

TOTAL_DISKS = 5  # More disks means a more difficult puzzle.

# Start with all disks on tower A:
SOLVED_TOWER = list(range(TOTAL_DISKS, 0, -1))

def main():
    """Runs a single game of The Tower of Hanoi."""
    print(
        """THE TOWER OF HANOI, by Al Sweigart al@inventwithpython.com

Move the tower of disks, one disk at a time, to another tower. Larger
disks cannot rest on top of a smaller disk.

More info at https://en.wikipedia.org/wiki/Tower_of_Hanoi
"""
    )

    """The towers dictionary has keys "A", "B", and "C" and values
    that are lists representing a tower of disks. The list contains
    integers representing disks of different sizes, and the start of
    the list is the bottom of the tower. For a game with 5 disks,
    the list [5, 4, 3, 2, 1] represents a completed tower. The blank
    list [] represents a tower of no disks. The list [1, 3] has a
    larger disk on top of a smaller disk and is an invalid
    configuration. The list [3, 1] is allowed since smaller disks
    can go on top of larger ones."""
    towers = {"A": copy.copy(SOLVED_TOWER), "B": [], "C": []}

    while True:  # Run a single turn on each iteration of this loop.
        # Display the towers and disks:
        displayTowers(towers)
```